

SASKATOON INTL, SK, Canada

WMO: 718660

Lat: 52.1736N

Lon: 106.7189W

Elev: 1654

StdP: 13.84

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -28.6	(c) -23.3	(d) -36.3	(e) 0.7	(f) -28.4	(g) -30.7	(h) 1.0	(i) -22.9	(j) 28.2	(k) 18.5	(l) 25.0	(m) 19.4	(n) 7.7	(o) 240	(p) 0.716

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 22.6	(c) 86.4	(d) 65.9	(e) 82.9	(f) 64.5	(g) 79.7	(h) 63.3	(i) 69.3	(j) 81.1	(k) 67.1	(l) 78.6	(m) 65.2	(n) 76.0	(o) 11.5	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.3	99.4	75.2	63.1	91.9	72.3	61.1	85.4	69.9	34.5	81.0	32.6	78.2	31.0	75.8	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
25.1	22.1	19.6	DB	-35.3	93.3	6.0	4.4	-39.7	96.4	-43.2	99.0	-46.5	101.5	-50.9	104.7	
			WB	-35.2	73.7	5.4	2.3	-39.1	75.3	-42.3	76.7	-45.4	77.9	-49.3	79.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.6	4.8	7.7	20.8	39.1	51.6	60.1	65.2	63.2	53.9	39.3	22.7	9.1
	DBStd	24.37	15.81	14.07	14.34	10.50	8.41	6.48	5.65	6.59	8.40	9.12	12.41	14.49
	HDD50	6414	1402	1183	904	348	82	5	0	1	53	347	820	1269
	HDD65	10535	1867	1603	1369	778	421	173	68	113	343	797	1269	1734
	CDD50	1529	0	0	0	20	132	309	471	411	170	16	0	0
	CDD65	175	0	0	0	0	6	27	73	59	10	0	0	0
	CDH74	2399	0	0	0	11	150	375	841	791	225	6	0	0
	CDH80	663	0	0	0	2	31	91	229	250	60	0	0	0
(14) Wind	WSAvg	9.6	9.4	9.3	10.1	10.9	11.0	9.9	8.8	8.7	9.4	9.6	9.4	9.0
(15) Precipitation	PrecAvg	14.3	0.7	0.6	0.7	0.8	1.8	2.6	2.3	1.5	1.3	0.7	0.6	0.7
	PrecMax	19.1	1.7	1.3	1.9	2.4	5.9	6.4	5.8	3.6	2.8	3.0	1.6	1.3
	PrecMin	9.3	0.2	0.0	0.2	0.1	0.3	0.5	0.6	0.1	0.2	0.0	0.0	0.2
	PrecStd	2.3	0.4	0.4	0.4	0.6	1.3	1.4	1.3	0.9	0.7	0.7	0.4	0.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.7	38.8	54.8	73.4	83.9	87.9	90.5	92.1	86.5	72.5	55.0	41.4
		MCWB	35.1	35.0	44.0	53.8	59.9	63.6	69.7	66.1	63.5	55.8	45.6	36.6
	2%	DB	35.7	34.7	47.4	67.7	78.2	82.2	85.7	86.2	80.5	64.5	47.0	36.9
		MCWB	33.0	32.2	40.7	50.3	57.3	62.6	68.0	65.8	61.0	51.1	40.7	33.5
	5%	DB	31.7	31.7	42.1	62.3	73.4	78.2	82.1	82.1	75.2	58.9	41.8	32.8
		MCWB	29.6	29.9	37.4	48.0	55.2	61.4	66.3	64.8	58.7	48.2	37.2	30.4
	10%	DB	26.6	27.9	37.9	56.8	69.0	74.5	78.8	78.2	70.1	53.8	38.1	28.3
		MCWB	25.1	26.5	34.6	45.2	53.0	59.6	65.1	63.2	56.1	45.4	34.7	26.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.6	35.7	45.6	54.9	62.5	69.6	73.9	70.5	65.5	56.7	46.2	36.9
		MCDB	38.4	38.2	53.2	70.2	76.1	80.6	85.9	83.4	81.1	70.7	54.0	40.9
	2%	WB	33.0	32.6	41.1	51.6	58.9	65.6	70.5	68.2	62.3	52.4	41.1	33.7
		MCDB	35.5	34.4	47.1	64.4	73.2	76.6	81.5	82.0	77.0	61.6	46.4	36.9
	5%	WB	29.6	29.8	37.6	48.9	56.5	63.4	68.2	66.1	59.7	49.2	37.6	30.4
		MCDB	31.8	31.6	41.9	61.0	71.1	74.2	78.9	78.2	72.5	57.4	41.0	32.8
	10%	WB	25.0	26.4	34.8	45.9	54.2	61.4	66.2	64.2	57.2	46.2	34.8	26.5
		MCDB	26.5	27.9	37.6	55.5	66.9	71.4	76.3	75.7	67.9	52.7	38.0	28.2
(35) Mean Daily Temperature Range	5% DB	MDBR	17.3	18.2	17.7	21.5	24.7	21.7	22.6	24.3	24.0	20.4	16.1	16.1
		MCDBR	19.2	19.6	20.7	30.4	31.6	28.0	27.7	30.6	33.3	29.1	21.0	18.9
		MCWBR	16.9	17.3	14.5	16.7	15.3	12.8	13.1	13.3	16.2	17.2	15.0	16.2
		MCDBR	19.1	19.0	19.9	28.7	27.9	23.3	24.3	26.2	29.9	27.0	19.7	18.8
	5% WB	MCWBR	16.9	16.9	14.3	16.4	14.1	12.0	12.8	12.6	15.2	16.4	14.5	16.2
(40) Clear-Sky Solar Irradiance	taub		0.251	0.266	0.296	0.343	0.357	0.377	0.377	0.379	0.324	0.282	0.253	0.219
	taud		2.285	2.275	2.349	2.339	2.346	2.338	2.342	2.330	2.469	2.560	2.473	2.375
	Ebn at Noon		243	273	286	283	284	278	276	268	274	269	247	244
	Edh at Noon		20	27	31	36	37	38	37	36	27	20	16	15
(44) All-Sky Solar Radiation	RadAvg		351	681	1153	1519	1870	1890	1990	1614	1172	688	346	253
(45)	RadStd		32	41	86	148	146	149	111	120	111	101	41	27

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page